

GOSAN SUWOLBONG, Korea, Republic of

WMO: 471850

Lat: 33.2939N Lon: 126.1628E Elev: 248 StdP: 14.56 Time Zone: 9.00 (E09) Period: 94-19 WBAN: 43263

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.6	34.4	20.0	15.1	38.0	21.9	16.6	37.7	56.3	37.3	52.2	37.5	33.3	340	0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.3	87.0	79.4	85.4	78.9	83.8	78.4	81.1	84.3	80.1	83.4	79.2	82.5	12.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.3	158.9	83.2	79.1	152.6	82.4	78.2	147.6	81.7	45.2	84.5	44.0	83.7	43.0	82.7	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
46.9	41.4	35.7		DB	30.1	89.8	2.9	2.0	28.0	91.2	26.3	92.4	24.6	93.5	22.5	94.9
				WB	27.1	81.9	2.0	1.6	25.7	83.1	24.5	84.0	23.3	84.9	21.9	86.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.5	42.7	43.9	48.8	56.3	62.9	69.5	77.2	79.9	73.8	65.7	56.4	47.2
	DBStd	13.36	5.36	5.68	5.28	4.55	3.64	3.03	4.03	3.24	3.50	4.45	5.75	5.61
	HDD50	645	234	185	85	6	0	0	0	0	0	0	11	124
	HDD65	2993	691	592	501	263	85	3	0	0	0	44	261	553
	CDD50	4461	9	13	49	194	400	585	843	928	714	487	203	36
	CDD65	1334	0	0	0	1	20	138	378	463	264	66	4	0
	CDH74	8454	0	0	0	1	7	155	2764	4379	1075	71	2	0
	CDH80	1728	0	0	0	0	0	2	524	1104	96	2	0	0
(14) Wind	WSAvg	15.6	22.2	20.5	18.2	15.0	12.3	10.0	11.7	11.9	11.9	14.5	17.5	21.4
(15) Precipitation	PrecAvg	52.0	1.9	2.2	3.4	4.2	4.9	6.7	7.5	8.8	5.5	2.3	2.8	1.7
	PrecMax	82.0	4.2	5.1	6.1	9.3	13.1	12.4	24.6	21.1	17.4	7.1	8.9	4.4
	PrecMin	30.6	0.4	0.4	0.5	1.2	2.1	1.6	0.4	2.4	0.5	0.1	0.1	0.3
	PrecStd	11.2	1.0	1.1	1.4	2.1	2.7	3.0	5.4	4.7	3.9	1.5	2.2	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.5	61.5	64.6	70.5	75.0	79.6	88.3	89.9	84.9	78.7	71.8	64.8
		MCWB	56.6	58.1	58.3	64.2	65.8	72.8	80.0	79.8	78.0	72.8	66.3	60.1
	2%	DB	56.8	58.8	61.4	67.0	72.5	77.3	86.2	87.8	82.6	75.9	69.5	60.8
		MCWB	52.5	55.6	57.2	61.3	65.3	71.8	79.7	79.4	76.5	69.6	64.5	54.9
	5%	DB	53.3	55.5	59.4	65.2	70.6	75.7	84.5	86.3	80.8	73.9	67.1	58.1
		MCWB	48.2	51.3	55.4	60.1	64.2	71.0	79.0	79.0	75.3	67.7	62.1	52.4
	10%	DB	50.7	52.8	57.1	63.3	68.9	74.4	82.9	85.0	79.1	71.9	64.7	55.5
		MCWB	46.1	48.4	52.8	58.4	63.4	70.3	78.3	78.7	74.0	65.7	59.9	50.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.4	59.3	60.6	66.3	70.1	76.8	83.1	83.0	80.0	75.3	68.8	62.1
		MCDB	59.8	60.9	62.5	67.9	71.6	78.1	85.9	86.3	82.6	77.2	70.8	63.8
	2%	WB	52.9	56.2	58.5	64.0	67.8	74.2	81.2	81.5	78.3	72.0	65.7	56.2
		MCDB	56.0	58.2	60.7	66.1	70.4	75.6	83.9	84.9	80.9	74.4	68.5	59.5
	5%	WB	49.1	52.2	56.1	61.8	66.3	72.9	80.0	80.5	77.0	69.3	63.0	53.1
		MCDB	52.5	54.7	58.6	63.9	68.9	74.3	82.8	84.2	79.8	72.4	66.2	57.3
	10%	WB	46.2	48.8	53.6	59.7	64.8	71.6	78.9	79.7	75.3	66.9	60.6	50.5
		MCDB	50.3	52.3	56.5	62.2	67.6	73.3	82.0	83.4	78.2	70.9	64.3	55.0
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	7.7	8.7	9.0	8.5	7.2	6.4	7.3	7.3	7.6	7.8	7.1
		MCDBR	11.3	12.0	12.1	11.1	10.8	8.9	8.0	9.0	9.1	10.2	10.6	11.0
		MCWBR	10.4	10.7	10.4	8.6	7.8	6.1	4.2	4.7	6.0	8.0	9.2	10.2
	5% WB	MCDBR	10.7	11.6	11.6	9.2	8.8	6.3	6.9	7.6	7.7	8.7	9.9	10.6
		MCWBR	10.9	11.9	11.3	8.4	7.3	5.2	4.2	4.6	5.7	8.3	9.8	10.9
(40) Clear-Sky Solar Irradiance	taub		0.465	0.504	0.553	0.561	0.568	0.598	0.582	0.536	0.492	0.466	0.464	0.460
	taud		1.947	1.886	1.787	1.824	1.861	1.880	1.970	2.109	2.168	2.101	2.026	1.964
	Ebn at Noon		224	231	231	237	237	229	232	241	246	240	224	217
	Edh at Noon		48	56	66	67	65	63	58	50	45	44	43	44
(44) All-Sky Solar Radiation	RadAvg		629	903	1297	1541	1650	1452	1595	1650	1369	1190	810	586
(45)	RadStd		75	115	105	127	177	176	222	166	128	119	91	59

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page